

LA RONGE, SK, Canada

WMO: 715860

Lat: 55.1464N

Lon: 105.2703W

Elev: 1242

StdP: 14.05

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.5	-27.9	-40.2	0.6	-31.7	-35.5	0.8	-27.6	20.8	6.7	18.5	4.0	2.8	280	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	82.1	63.9	78.9	62.5	76.0	61.0	66.4	77.6	64.7	75.0	63.1	72.5	9.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	88.3	70.4	60.9	83.5	68.7	59.3	78.8	67.2	31.8	77.7	30.5	75.6	29.2	72.3	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
18.9	17.4	15.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-39.0	89.0	5.1	3.3	-42.7	91.4	-45.6	93.3	-48.5	95.1	-52.2	97.5		
			-39.1	69.9	5.1	1.5	-42.8	71.0	-45.7	71.8	-48.6	72.7	-52.2	73.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	33.0	-0.9	3.5	16.7	33.7	47.0	58.4	63.8	61.3	51.0	37.1	18.9	3.7	(6)
(7)		DBStd	25.48	16.10	14.21	14.85	11.30	8.78	6.61	5.38	6.36	8.09	8.29	12.06	15.40	(7)
(8)		HDD50	7442	1579	1301	1032	495	163	11	0	3	84	405	933	1436	(8)
(9)		HDD65	11777	2044	1721	1497	939	558	214	86	150	421	863	1383	1901	(9)
(10)		CDD50	1243	0	0	0	7	71	262	429	352	115	7	0	0	(10)
(11)		CDD65	104	0	0	0	0	1	16	50	34	3	0	0	0	(11)
(12)		CDH74	1072	0	0	0	1	47	188	472	319	45	0	0	0	(12)
(13)		CDH80	197	0	0	0	0	8	32	86	65	6	0	0	0	(13)
(14)	Wind	WSAvg	6.8	6.1	6.2	6.9	7.5	7.6	7.2	6.7	6.6	7.1	7.3	6.8	5.9	(14)
(15)	Precipitation	PrecAvg	19.7	0.7	0.6	0.8	1.2	1.7	3.0	3.4	2.5	2.3	1.5	1.1	0.9	(15)
(16)		PrecMax	29.8	1.5	1.4	2.3	3.0	5.1	11.7	7.9	5.9	5.9	2.9	2.0	1.8	(16)
(17)		PrecMin	12.6	0.2	0.1	0.2	0.0	0.1	0.4	1.6	0.4	0.6	0.4	0.1	0.3	(17)
(18)		PrecStd	3.5	0.4	0.3	0.5	0.8	1.3	1.9	1.5	1.3	1.3	0.6	0.5	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.0	41.6	51.8	69.0	79.6	84.0	86.4	86.0	79.0	65.7	46.3	37.5	(19)
MCWB			35.3	35.0	40.3	50.4	57.3	62.9	66.4	66.7	61.2	53.5	39.2	33.2	(20)	
2%		DB	32.8	34.4	45.7	62.0	73.2	79.1	82.2	81.2	73.8	58.5	41.5	32.4	(21)	
		MCWB	30.2	30.7	37.0	47.3	54.7	60.9	64.7	64.2	59.6	48.9	36.6	30.1	(22)	
5%		DB	27.2	29.5	41.0	56.1	68.8	75.3	79.1	77.5	68.7	53.6	37.3	28.4	(23)	
		MCWB	25.5	27.0	34.5	43.7	52.5	59.1	63.2	63.0	57.1	46.2	33.9	26.7	(24)	
10%		DB	22.4	25.2	36.6	50.7	64.3	72.0	76.1	73.8	64.4	49.3	33.9	24.3	(25)	
		MCWB	21.3	23.5	32.3	40.8	50.2	57.5	62.0	61.3	54.9	43.9	31.6	23.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.9	36.0	41.1	51.6	59.5	66.7	69.3	68.7	63.6	54.7	40.3	34.6	(27)
MCDB			39.3	41.2	50.7	67.3	74.6	78.3	81.4	81.3	73.7	64.2	44.9	37.0	(28)	
2%		WB	30.1	31.0	37.5	47.3	56.4	63.7	66.8	66.1	60.7	50.0	37.0	30.3	(29)	
		MCDB	32.6	34.2	44.6	60.5	69.3	73.7	77.9	77.3	70.7	56.1	40.4	32.4	(30)	
5%		WB	25.6	27.2	34.9	44.3	53.7	61.6	65.2	64.4	58.3	46.8	34.1	26.8	(31)	
		MCDB	27.1	29.5	40.2	55.2	66.7	71.2	75.5	74.6	67.5	52.6	36.8	28.3	(32)	
10%		WB	21.3	23.3	32.4	41.2	51.1	59.5	63.6	62.6	55.7	44.1	31.8	23.1	(33)	
		MCDB	22.6	25.2	36.5	49.9	62.4	68.5	73.2	71.5	63.2	48.7	33.9	24.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.4	19.9	21.7	21.3	22.2	19.7	19.7	19.1	17.9	13.6	12.5	14.6	(35)
MCDBR			19.7	21.7	23.8	28.6	30.5	25.9	24.9	24.9	25.2	21.5	15.4	16.8	(36)	
MCWBR			17.4	18.3	16.4	16.7	15.4	11.8	11.1	11.8	13.6	13.5	11.4	15.1	(37)	
5% WB		MCDBR	19.3	21.0	21.6	27.5	26.9	20.5	21.9	22.0	23.7	20.3	14.1	16.3	(38)	
		MCWBR	17.2	17.8	15.2	16.4	14.3	10.5	10.6	11.2	13.5	13.2	11.0	14.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.238	0.245	0.272	0.314	0.336	0.365	0.376	0.357	0.312	0.288	0.243	0.214	(40)	
(41)		taud	2.288	2.348	2.363	2.453	2.474	2.425	2.400	2.477	2.590	2.577	2.434	2.308	(41)	
(42)		Ebn at Noon	230	274	290	291	288	279	273	273	274	257	232	218	(42)	
(43)		Edh at Noon	16	23	29	31	32	34	34	30	23	19	14	13	(43)	
(44)	All-Sky Solar Radiation	RadAvg	254	585	1074	1550	1754	1774	1741	1454	986	531	239	164	(44)	
(45)		RadStd	27	36	83	106	166	119	107	128	112	80	34	21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page